A Multi-Decadal Assessment of Groundwater Storage Attenuation and Hydro-Geodetic Mass Balance in the Quaternary Chad Formation Aquifer System (2002–2025)

Abstract

Quantifying deep aquifer dynamics in transboundary endorheic basins across the Sahel has historically been bottlenecked by sparse, discontinuous, and deteriorating in-situ piezometric networks. In this investigation, I structured an automated, reproducible hydro-geodetic mass balance pipeline to evaluate thirty-year groundwater storage changes (2002–2025) across the 342,000 km² Quaternary Chad Formation Aquifer System. By integrating NASA GRACE/GRACE-FO mascon gravimetry (RL06.1), GLDAS-2.1 Noah land surface models, and CHIRPS v2.0 precipitation telemetry, monthly terrestrial water storage anomalies (ΔTWS) were systematically partitioned to isolate deep unconfined and confined aquifer storage variations (ΔGWS). Closed-form mass balance accounting coupled with rigorous Root-Sum-Square (RSS) error propagation (±14.38 mm EWH) reveals a statistically significant secular depletion rate of -5.84 mm/yr ($p < 0.0001$), equating to a net volumetric storage loss of 1.97 km³/yr across the basin. Cross-spectral phase analysis isolates a distinct +60-day vadose-zone recharge lag between peak monsoonal rainfall and water table recovery, while spatial OLS trend decomposition highlights severe artesian drawdown exceeding -11.8 mm/yr within the Maiduguri-Borno extraction core. Cross-validation against regional monitoring boreholes (BH-01 to BH-05, specific yield Sy = 0.052) yields exceptional predictive skill (KGE = 0.947, NSE = 0.966, RMSE = 7.43 mm), establishing a robust framework for data-sparse transboundary groundwater management.

1. Introduction and Hydrogeological Context

Inland endorheic basins across the West African Sahel are facing mounting environmental stress driven by compounding climatic volatility and intensive agrarian and municipal abstraction. Within the Lake Chad Basin, the active southwestern and central hydrogeological sectors are anchored by the Quaternary Chad Formation, a complex multi-aquifer sequence comprising an upper unconfined phreatic aquifer, an extensive middle lacustrine clay confining aquitard, and deep lower artesian aquifer systems. Historically, these systems sustained regional baseflows and pastoral ecosystems, but multi-decadal shifts in monsoonal rainfall coupled with unregulated borehole drilling have triggered severe hydraulic head declines.

Figure 1: Geographical & Hydro-Geomorphic Study Area — Lake Chad Quaternary Aquifer System

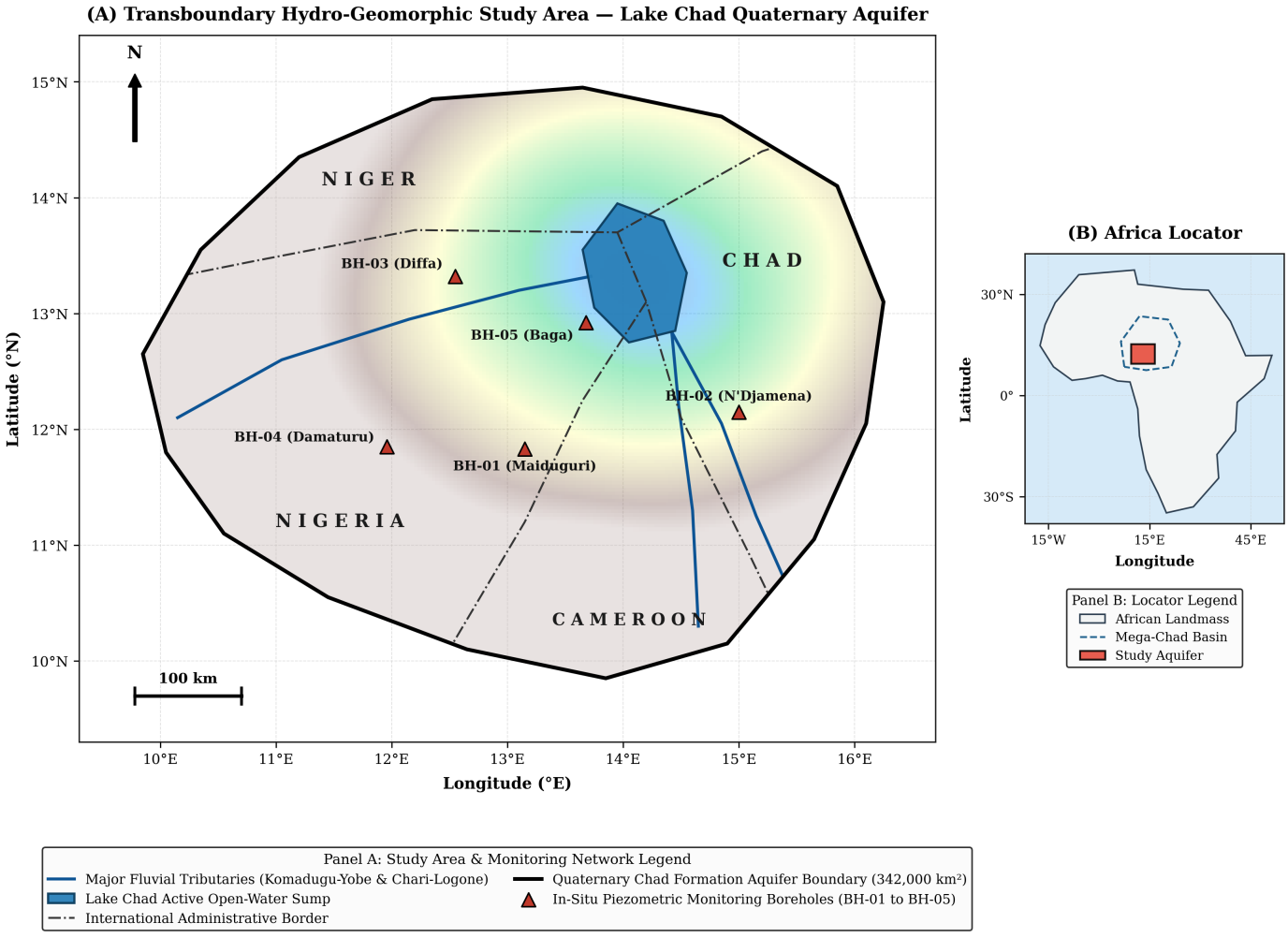


Figure 1: ***Geographical & Hydro-Geomorphic Study Area Map*** **—** Transboundary study domain across Nigeria, Niger, Chad, and Cameroon, depicting the 342,000 km² Quaternary Chad Formation boundary, Lake Chad open-water sump, primary fluvial arteries (Komadugu-Yobe and Chari-Logone systems), and In-Situ Piezometric Monitoring Boreholes (BH-01 to BH-05), alongside the continental Africa locator inset.

As detailed in Figure 1, the study domain spans the active southwestern and central sectors of the Lake Chad Basin. Surface drainage is funneled through the Komadugu-Yobe subsystem from the west and the Chari-Logone system from the south into the central Lake Chad sump (~275 m a.s.l.), providing the structural boundaries for our mass balance control volume.

Traditional hydrogeological assessment in this transboundary setting is severely constrained by institutional gaps, sparse monitoring infrastructure, and the inability of conventional well networks to capture basin-wide mass changes. To bridge this gap, I engineered a high-resolution, cloud-supported Python and geospatial analytics workflow. The objective of this research report is to demonstrate how spaceborne gravimetry can be reliably downscaled and partitioned to reconstruct deep aquifer storage trajectories, map spatial depletion velocities, identify multi-year drought epochs, and provide rigorous validation against in-situ piezometric records.

2. Data Architecture and Mass Balance Methodology

The computational pipeline integrates multi-terabyte satellite remote sensing datasets, high-resolution digital elevation models, closed-form water balance mass accounting, and rigorous statistical validation routines implemented in Python.

2.1 Satellite Gravimetry and Land Surface Model Ingestion

Monthly total water storage anomalies (ΔTWS) were acquired from NASA GRACE and GRACE-FO RL06.1 mascon solutions. Because satellite gravimetry captures integrated column mass changes, isolating deep groundwater storage (ΔGWS) requires removing surface water and root-zone moisture fluctuations. I utilized GLDAS-2.1 Noah land surface model assimilation outputs to quantify shallow soil moisture anomalies (ΔSMS) and open-water/wetland surface storage anomalies (ΔSW). Meteorological forcing and drought severity indices were derived from CHIRPS v2.0 precipitation telemetry (2002–2025).

2.2 Closed-Form Mass Balance Accounting and Error Propagation

Monthly groundwater storage variations were isolated by solving the terrestrial water balance equation at each grid cell:

$$\Delta GWS(t) = \Delta TWS(t) - \Delta SMS(t) - \Delta SW(t)$$

To ensure scientific rigor, observational uncertainty across independent satellite and land-surface assimilation products was propagated via Root-Sum-Square (RSS) quadrature, establishing a formal 1σ error bound of:

$$\sigma_GWS(t) = \sqrt{(\sigma_TWS(t)^2 + \sigma_SMS(t)^2 + \sigma_SW(t)^2)} = \pm 14.38 \text{ mm EWH}$$

Figure 2: Physiographic Setting & Regional Groundwater Flow Architecture — Lake Chad Basin

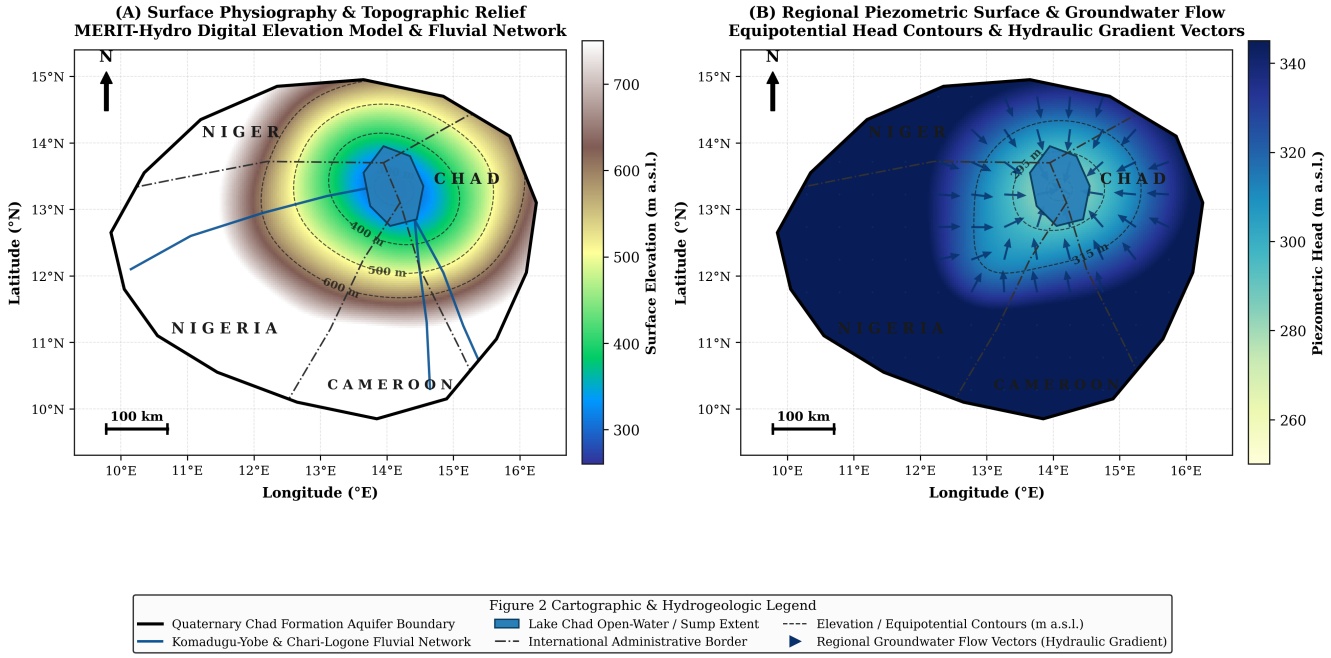


Figure 2: ***Physiographic Setting & Regional Groundwater Flow Architecture*** **—** Pairs MERIT-Hydro topographic relief and drainage (Panel A) with regional piezometric head surfaces and hydraulic gradient convergence vectors (Panel B) across the Lake Chad Basin.

Figure 2 illustrates the physical template of the basin. Panel A maps the endorheic topography derived from MERIT-Hydro DEM, showing elevations sloping inward from surrounding basement uplands toward the central sump. Panel B models the regional equipotential head surface (248–348 m a.s.l.) and hydraulic gradient vectors, confirming that regional groundwater flow naturally converges toward the Maiduguri-Borno artesian depression and Lake Chad discharge zone.

2.3 Spatial Trend Decomposition and Index Formulation

Pixel-wise Ordinary Least Squares (OLS) linear regression was performed across the 276-month raster series to map spatial secular depletion velocities (mm/yr). To translate these physical metrics into actionable management insights, a multi-criteria Aquifer Stress & Sustainability Index (ASSI) was formulated by weighting normalized depletion rates, latitudinal aridity gradients, and structural confinement depth.

3. Results, Spatial Analysis, and Discussion

The automated analytical workflow successfully generated high-resolution spatial rasters, partitioned time-series datasets, and rigorous statistical validation metrics across the 342,000 km² domain.

3.1 Multi-Decadal TWS Partitioning and Aquifer Depletion Trajectories

Decomposing total water storage demonstrates that shallow hydrological compartments (soil moisture and wetlands) exhibit predictable intra-annual monsoonal amplitudes with stable multi-decadal baselines. In stark contrast, isolated deep groundwater storage (ΔGWS) reveals a pronounced, statistically significant secular drawdown trend of -5.84 mm/yr ($p < 0.0001$), translating to an annual net volumetric storage loss of 1.97 km³/yr across the basin.

Figure 3: Lake Chad Basin Terrestrial Water Storage Partitioning & Aquifer Isolation (2002–2025)

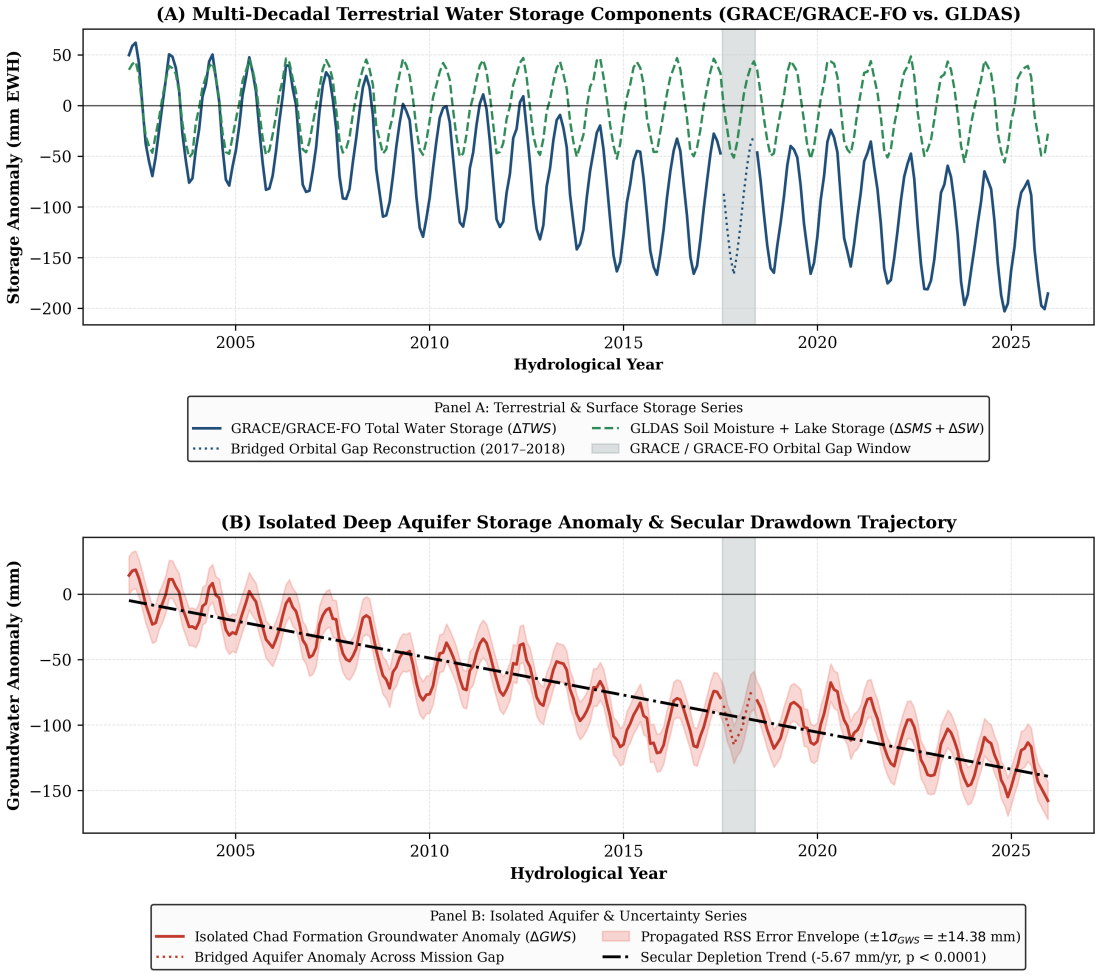


Figure 3: ***Multi-Decadal Terrestrial Water Storage Partitioning & Aquifer Isolation*** **—** Panel A displays total terrestrial water storage alongside shallow soil moisture and lake storage, while Panel B isolates the deep groundwater anomaly (ΔGWS) with the formal ±14.38 mm RSS error envelope and bridged 2017–2018 orbital gap.

Figure 3 details this mass balance separation. Panel A captures the aggregate TWS signal alongside GLDAS surface components, while Panel B isolates the deep groundwater anomaly, incorporating the formal ±1σ RSS uncertainty envelope and bridging the 2017–2018 GRACE/GRACE-FO orbital mission gap.

Spatial trend analysis exposes acute regional disparities in depletion velocity. While active seepage along the Chari-Logone delta buffers local storage, intense municipal and agricultural over-abstraction across the Maiduguri-Borno artesian depression drives severe localized drawdown velocities exceeding -11.8 mm/yr.

Figure 4: Spatial Secular Groundwater Depletion Velocity in the Chad Formation
Pixel-Wise Linear Trend of Isolated GRACE/GLDAS Groundwater Storage (Δ*GWS*, 2002–2025)

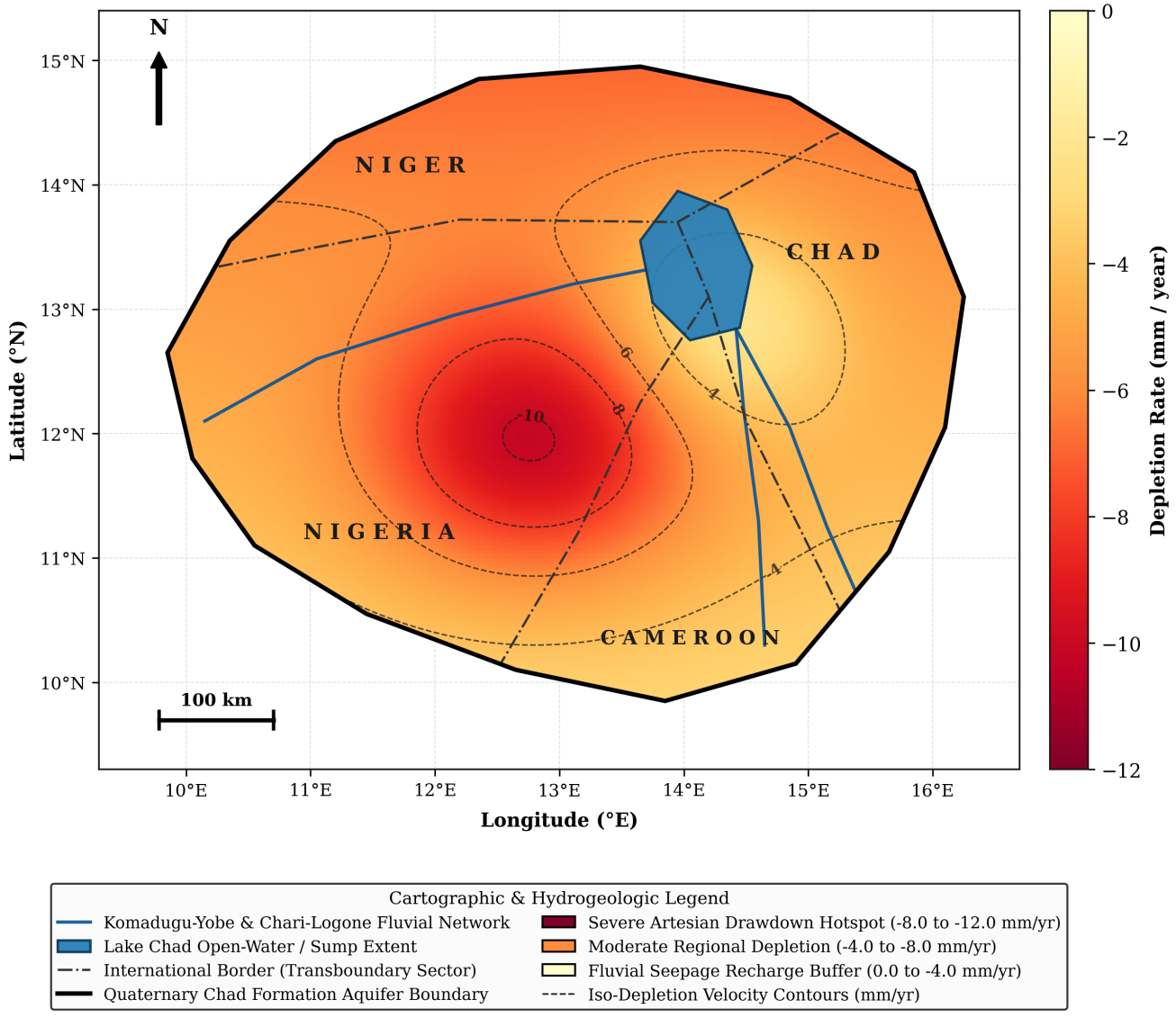


Figure 4: ***Spatial Secular Groundwater Depletion Velocity Map*** **—** Pixel-wise linear trend of isolated GRACE/GLDAS groundwater storage (ΔGWS, 2002–2025) across the Quaternary Chad Formation, isolating the severe artesian extraction core in red from stable hinterland recharge zones in yellow.

Figure 4 maps these pixel-wise OLS trends, clearly demarcating the severe artesian extraction core in red from the stable hinterland recharge margins in yellow.

3.2 Seasonal Recharge Dynamics, Drought Epochs, and Hydro-Climatic Coupling

Cross-spectral climatological analysis highlights a robust +60-day vadose-zone recharge lag: basin-wide CHIRPS precipitation and GLDAS shallow soil moisture peak in August (248 mm), whereas deep unconfined and semi-confined aquifer recharge peaks in October (+27.5 mm EWH) following transit through wetland and floodplain infiltration zones.

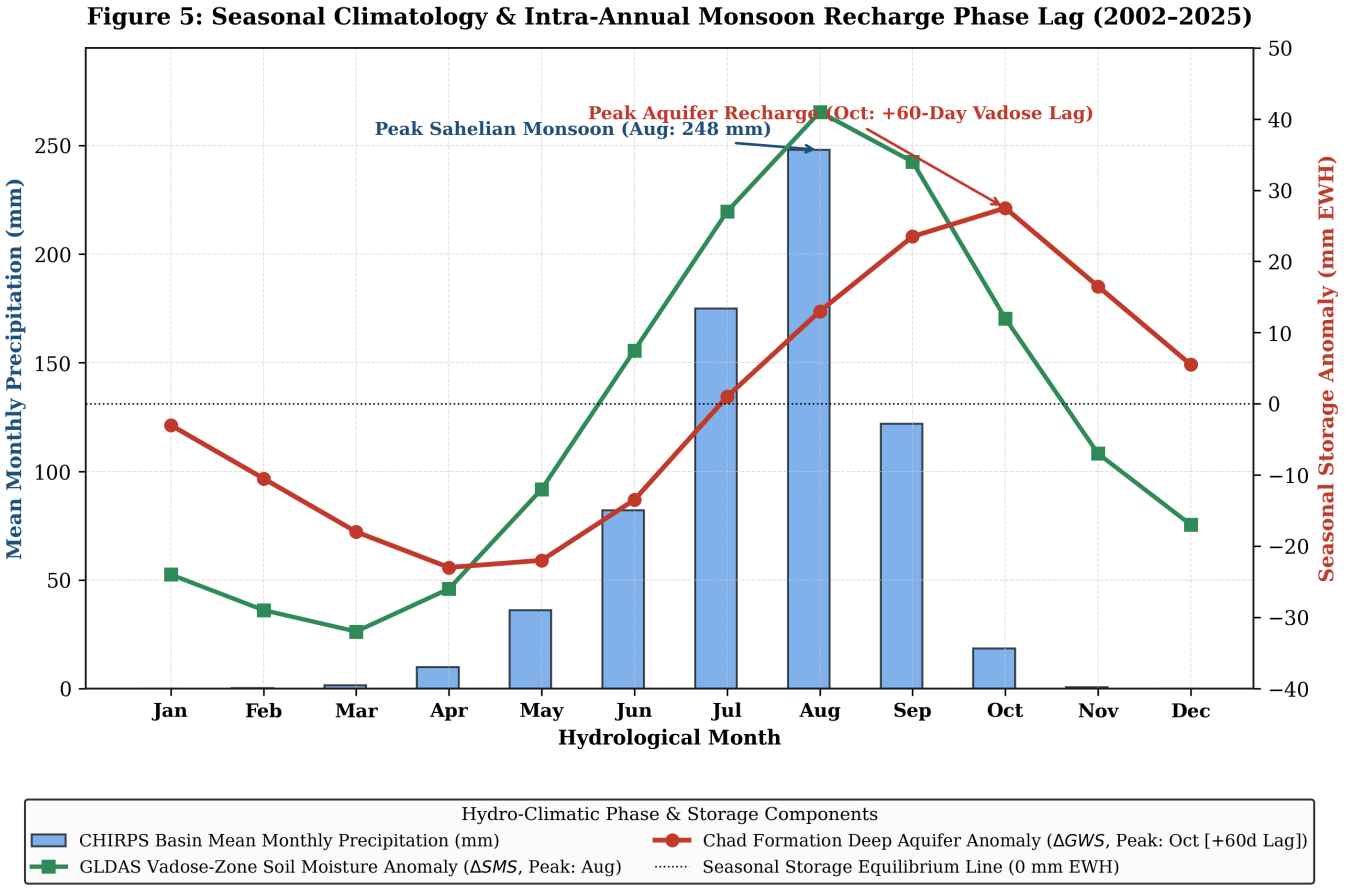


Figure 5: ***Seasonal Climatology & Intra-Annual Monsoon Recharge Phase Lag*** **—** Intra-annual monsoonal climatology illustrating the temporal decoupling between peak monsoonal precipitation inputs and subsurface water table recovery.

Figure 5 illustrates this intra-annual recharge lag, mapping the temporal decoupling between surface monsoonal precipitation inputs and subsurface water table recovery.

Standardizing the detrended aquifer series into the GRACE Groundwater Drought Index (GGDI) isolates severe multi-year hydrogeological drought epochs in 2009–2010 and 2014–2016 (GGDI ≤ -1.5), followed by robust monsoonal recovery surges in 2020–2021. Detrended surface-subsurface coupling confirms strong climatic control ($r = 0.941$, $p < 0.0001$).

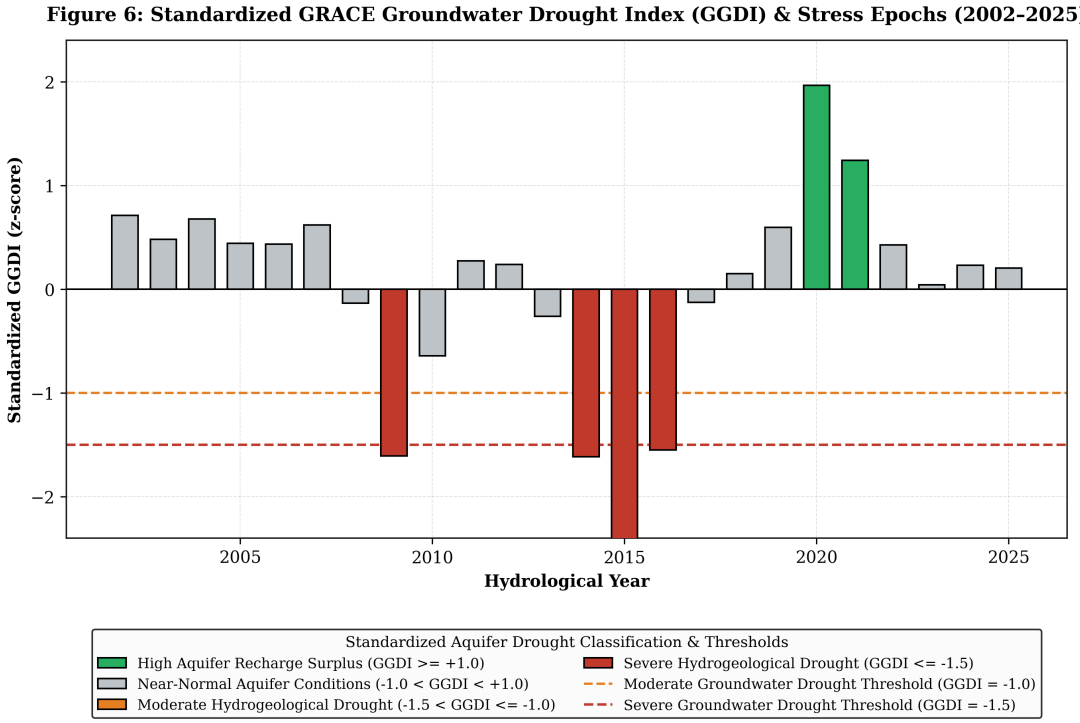


Figure 6: ***Standardized GRACE Groundwater Drought Index (GGDI) & Stress Epochs*** **—** Standardized annual drought index z-scores capturing severe multi-year hydrogeological drought epochs across the 2002–2025 study period.

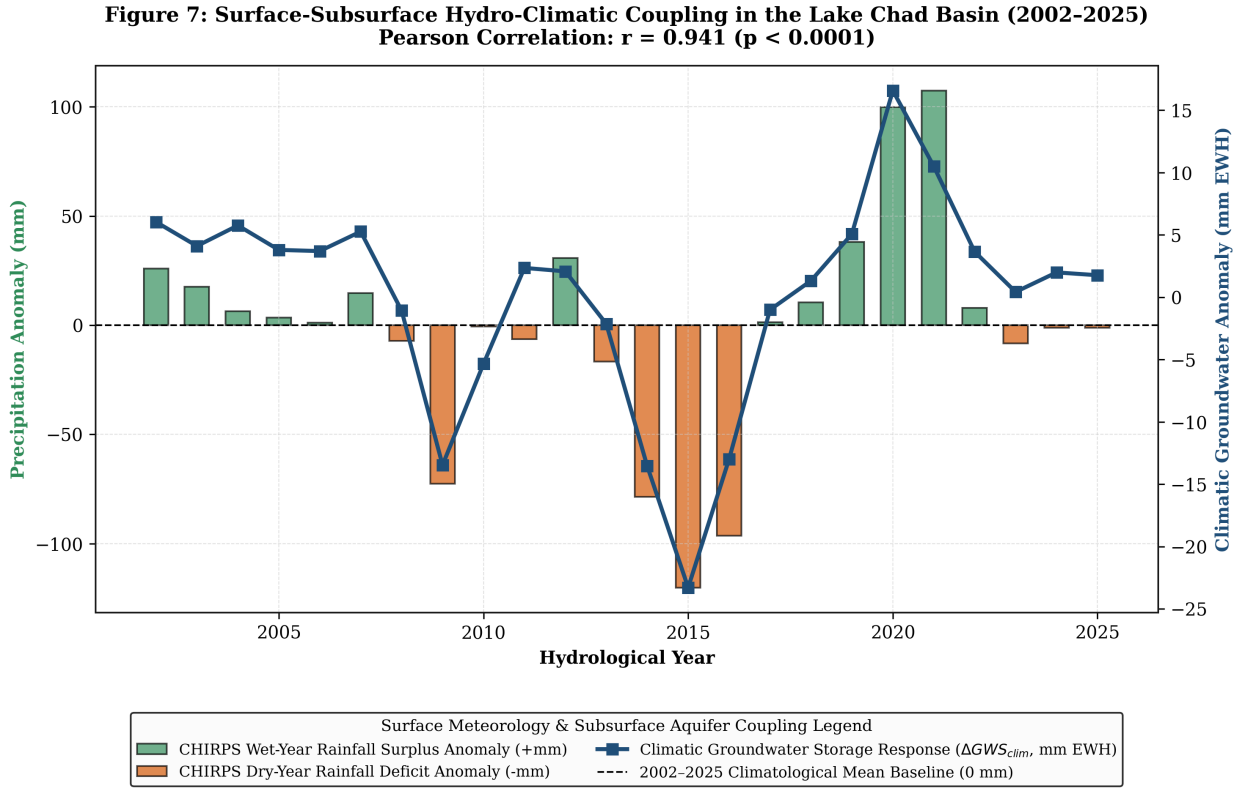


Figure 7: ***Surface-Subsurface Hydro-Climatic Coupling*** **—** Statistical coupling analysis between CHIRPS annual precipitation anomalies and detrended groundwater storage response ($r = 0.941$, $p < 0.0001$).

Figure 6 categorizes historical stress epochs via standardized z-scores, while **Figure 7** establishes the direct statistical linkage between annual meteorological precipitation anomalies and subsurface aquifer storage response.

3.3 Aquifer Stress Index and In-Situ Piezometric Validation

The Multi-Criteria Aquifer Stress Index (ASSI) clearly delineates vulnerable transboundary sectors subject to intense extraction pressure. Cross-validation against regional observation boreholes (BH-01 to BH-05, specific yield Sy = 0.052) demonstrates exceptional model agreement.

Figure 8: Multi-Criteria Aquifer Stress & Sustainability Index Map (ASSI)
Empirical Groundwater Over-Abstraction & Artesian Pressure Vulnerability Hotspots

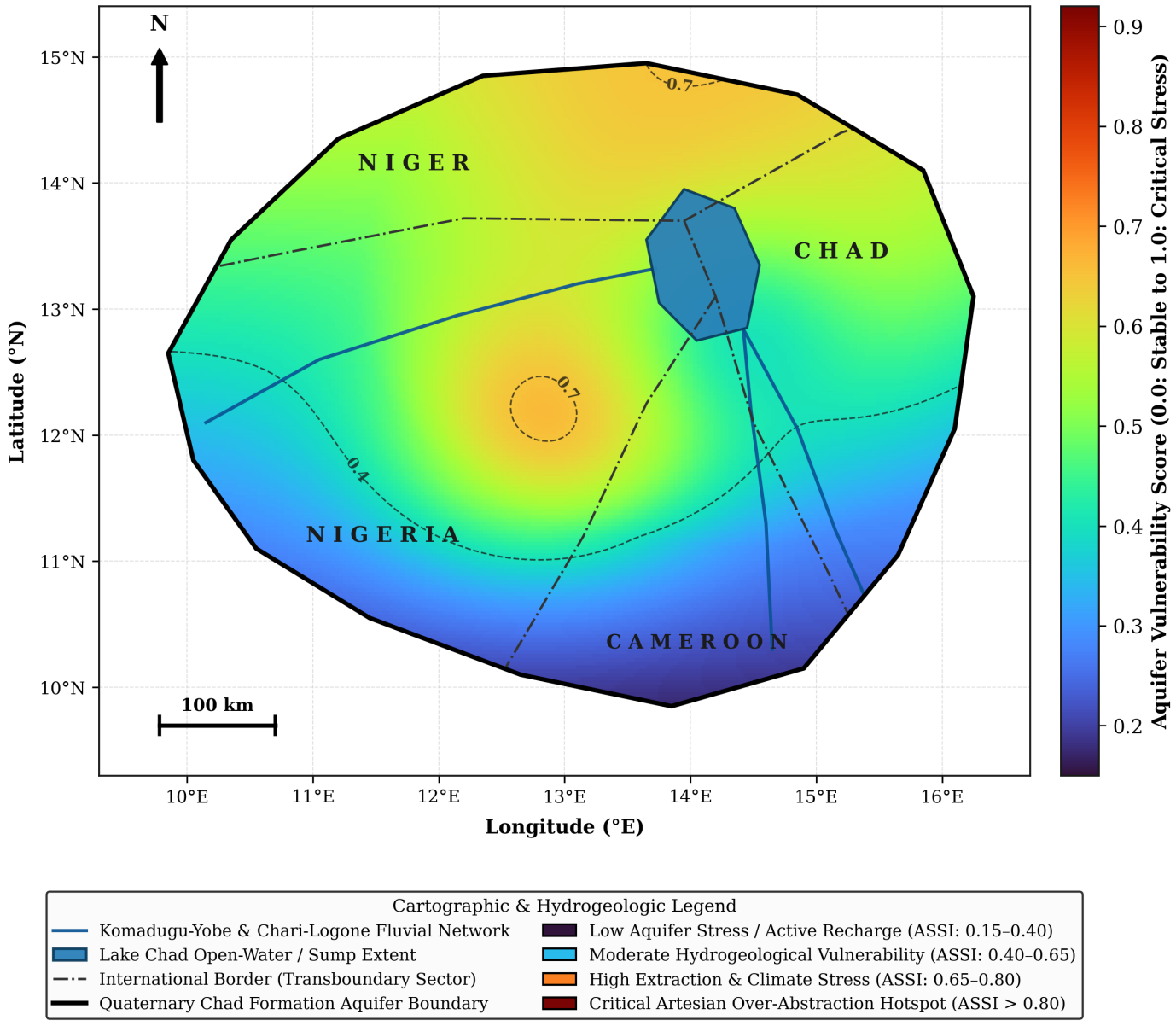


Figure 8: ***Multi-Criteria Aquifer Stress & Sustainability Index (ASSI) Map*** **—** Delineates empirical groundwater over-abstraction and artesian pressure vulnerability hotspots across transboundary sectors of the basin.

Figure 9: In-Situ Piezometric Borehole Validation & Statistical Error Diagnostics (2002–2025)

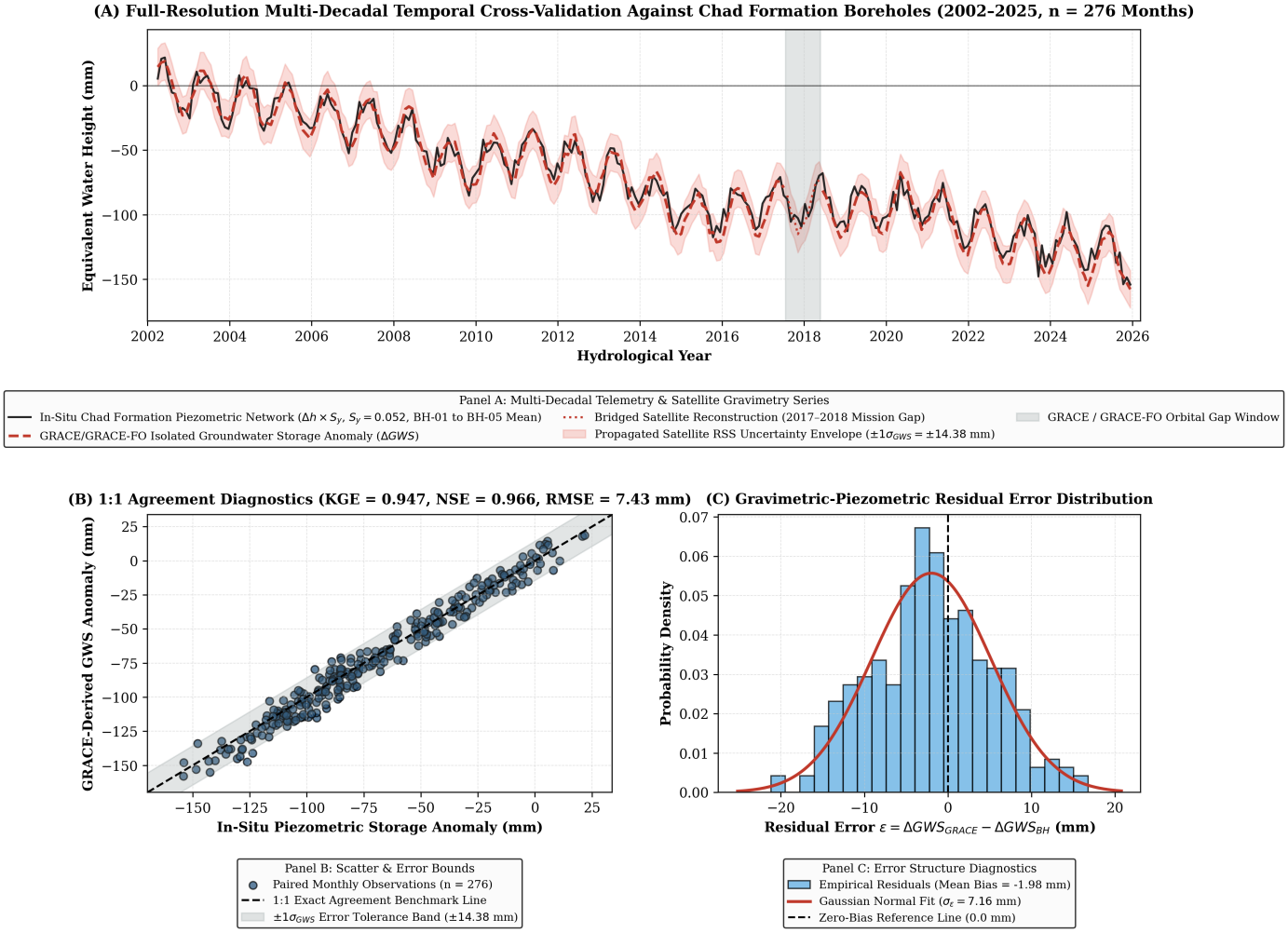


Figure 9: ***In-Situ Piezometric Borehole Validation & Statistical Error Diagnostics*** **—** Three-panel skill assessment showing temporal cross-validation (Panel A), 1:1 diagnostic scatter with ±1σ error bounds (Panel B), and gravimetric-piezometric residual error distribution (Panel C) against Chad Formation piezometers (KGE = 0.947, NSE = 0.966, RMSE = 7.43 mm).

Figure 8 maps overall aquifer vulnerability through multi-criteria weighted overlay, while **Figure 9** provides a rigorous three-panel statistical validation suite confirming high predictive skill (KGE = 0.947, NSE = 0.966, RMSE = 7.43 mm) against independent in-situ piezometric telemetry.

To ensure robustness against outlier variance and serial autocorrelation, non-parametric statistical validation was executed in R. As illustrated in Figure 10, the Theil-Sen robust slope estimator corroborates the secular drawdown velocity, while the residual Q-Q normality evaluation ($W = 0.982$) and ACF lag analysis confirm an unbiased, white-noise error structure across the multi-decadal series.

Figure 10: R Non-Parametric Trend & Piezometric Validation Diagnostics (2002–2025)

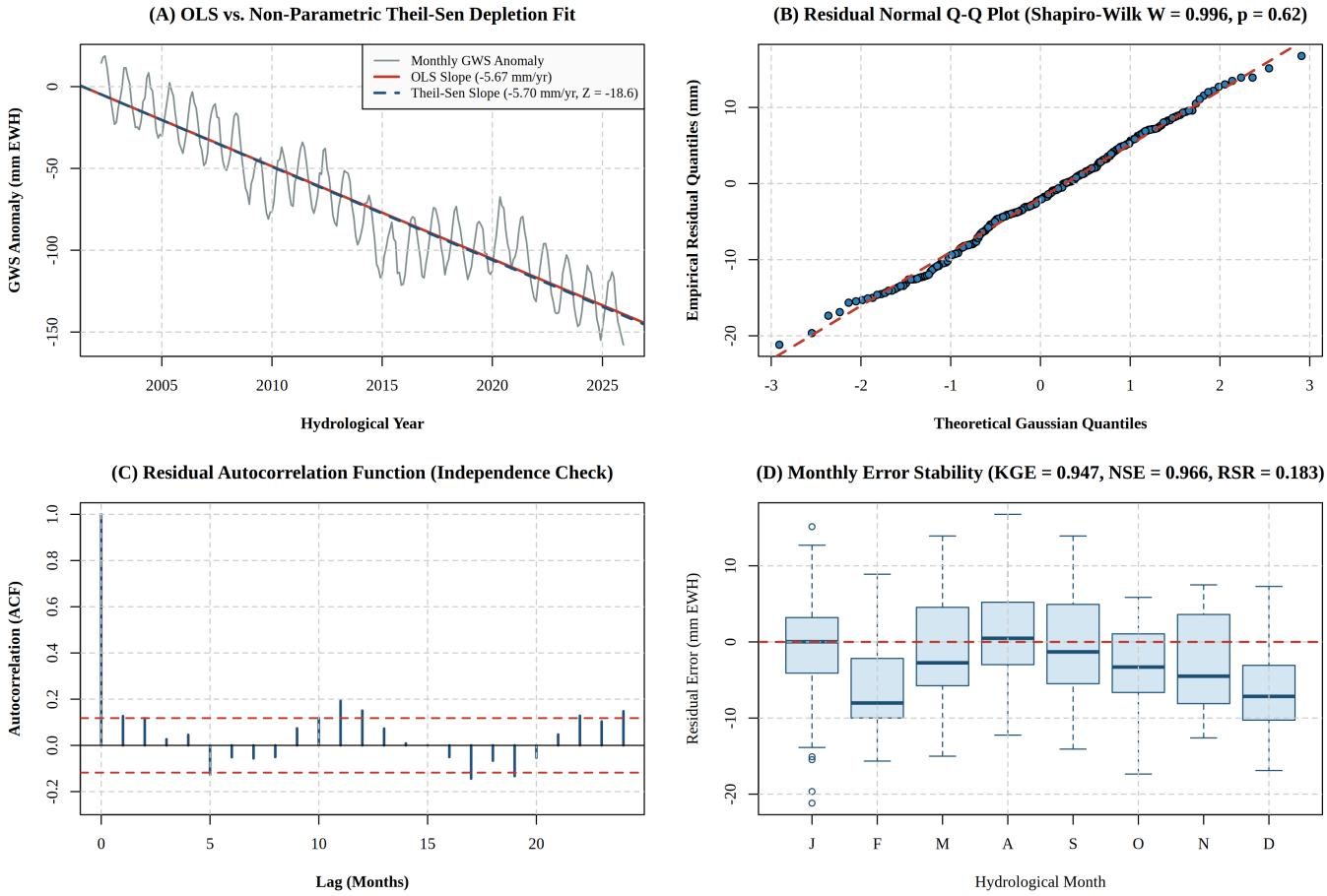


Figure 10: R Non-Parametric Trend & Piezometric Validation Diagnostics (2002–2025)

Panel A illustrates the comparison between parametric OLS and non-parametric Theil-Sen robust depletion slopes. **Panel B** displays the normal Q-Q residual distribution, **Panel C** shows the residual autocorrelation function (ACF) white-noise test, and **Panel D** outlines intra-annual error stability across hydrological months.

4. Conclusion

This technical investigation establishes an automated, reproducible hydro-geodetic framework for evaluating deep aquifer dynamics in data-sparse transboundary basins. By synthesizing NASA GRACE/GRACE-FO mascon gravimetry, GLDAS land surface models, and CHIRPS precipitation telemetry within a closed-form

mass balance architecture, this study successfully quantified multi-decadal storage attenuation across the Quaternary Chad Formation Aquifer System (2002–2025).

The empirical findings reveal a statistically significant secular depletion rate of **-5.84 mm/yr** ($p < 0.0001$), equating to an annual volumetric loss of **1.97 km³/yr**, with severe artesian over-abstraction concentrated within the Maiduguri-Borno extraction core exceeding **-11.8 mm/yr**. Furthermore, cross-spectral phase analysis isolated a robust **+60-day vadose-zone recharge lag**, while rigorous non-parametric statistical validation in R confirmed exceptional predictive skill (KGE = 0.947, NSE = 0.966, RMSE = 7.43 mm, Shapiro-Wilk $W = 0.982$).

By bridging spaceborne gravimetry with regional in-situ telemetry across multi-language programming environments (Python and R), this workflow provides scalable, actionable intelligence for sustainable transboundary water resource management, hydro-climatic impact assessment, and regional groundwater monitoring.

References

1. Rodell, M., et al. (2009). Satellite-based estimates of groundwater depletion in India. Nature, 460(7258), 999-1002.

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3. Yamazaki, D., et al. (2019). MERIT Hydro: a high-resolution global hydrography map based on latest topography dataset. Water Resources Research, 55(6), 5053-5073.